

Practical 1

A]Create a Simple Calculator using Servlet Application.

Code:

Index.html

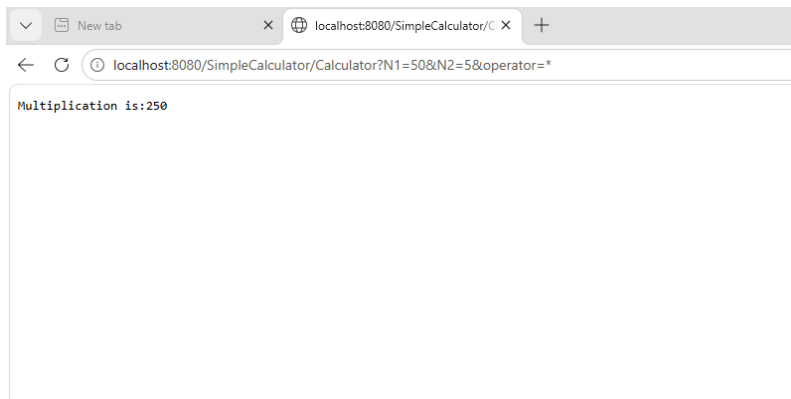
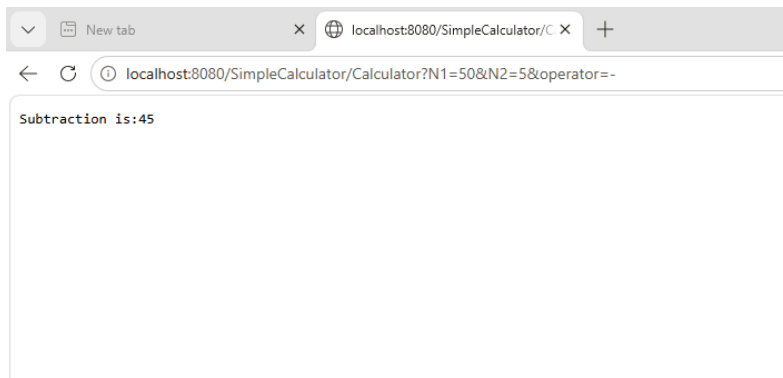
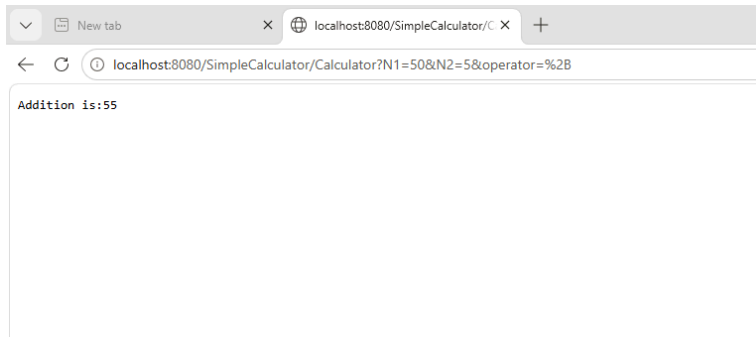
```
<html>
  <head>
    <title>Servlet Calculator</title>
  </head>
  <body>
    <form action= "Calculator" method="get">
      Enter number 1:<input type="text" name="N1"><br><br>
      Enter number 1:<input type="text" name="N2"><br><br>
      Select operator: <select name="operator">
        <option value="+">+</option>
        <option value="-">-</option>
        <option value="*">*</option>
        <option value="/">/</option>
      </select><br><br><br>
      <input type="submit" value="Calculate">
      <input type="submit" value="Reset">
    </form>
  </body>
</html>
```

Calculator.Java

```
import java.io.*;
import javax.servlet.*;

public class Calculator extends GenericServlet {
    public void service(ServletRequest req, ServletResponse res) throws ServletException,
    IOException {
        PrintWriter out = res.getWriter();
        int a = Integer.parseInt(req.getParameter("N1"));
        int b = Integer.parseInt(req.getParameter("N2"));
        String OP=req.getParameter("operator");
        if(OP.equals("+")){
            out.println("Addition is: "+(a+b));
        }else if(OP.equals("-")){
            out.println("Subtraction is: "+(a-b));
        }else if(OP.equals("*")){
            out.println("Multiplication is: "+(a*b));
        }else{
            out.println("Division is: "+(a/b));
        }
    }
}
```

```
}  
}
```



Division is:10

B] Create a Servlet for a login page. If the username and password are correct then if it says “Hello <username>” else message “Login failed”.

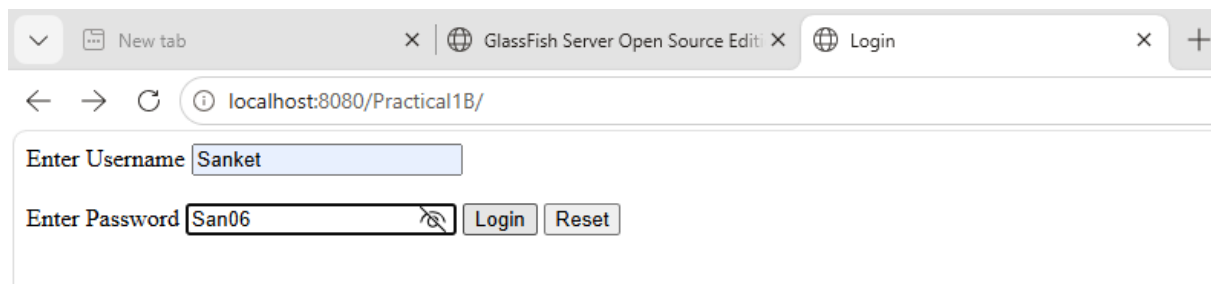
Code:

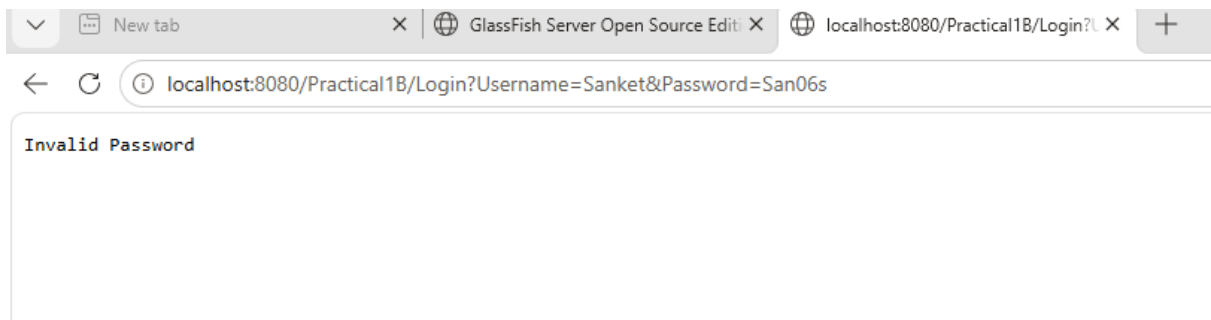
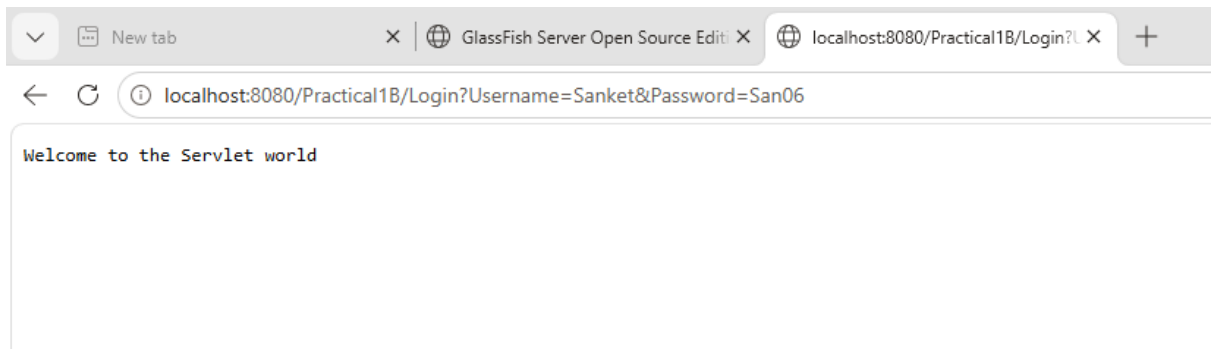
Index.html

```
<html>
  <head>
    <title>Login </title>
  </head>
  <body>
    <form action="Login" method="get">
      Enter Username <input type="text" name="Username"><br><br>
      Enter Password <input type="password" name="Password">
      <input type="Submit" value="Login">
      <input type="Reset" value="Reset">
    </form>
  </body>
</html>
```

Login.java

```
import java.io.*;
import javax.servlet.*;
public class Login extends GenericServlet{
    public void service(ServletRequest req, ServletResponse res) throws ServletException,
IOException {
    PrintWriter out = res.getWriter();
    String Username = req.getParameter("Username");
    String Password = req.getParameter("Password");
    if (Username.equals("Sanket") && Password.equals("San06")){
        out.println("Welcome to the Servlet world");
    }else{
        out.println("Invalid Password");
    }
}
}
```





C]Create a registration servlet in Java using JDBC. Accept the details such as *Username, Password, Email, and Country* from the user using HTML Form and store the registration details in the database.

Code:

Index.html

```
<html>
  <head>
    <title>Registration Servlet</title>
  </head>
  <body>
    <form action="Register" method="get">
      Enter Username:<input type="text" name="n1"><br>
      Enter password:<input type="text" name="n2"><br>
      Enter Email:<input type="text" name="n3"><br>
      Enter Country Name:<input type="text" name="n4"><br>
      <input type="submit" value="Register">
      <input type="reset" value="Reset">

    </form>
  </body>
</html>
```

Register.java

```
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

@WebServlet(urlPatterns = {"/register"})

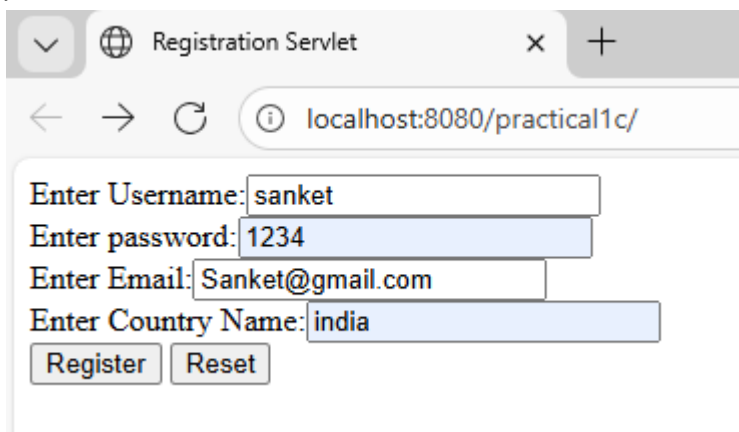
public class Register extends HttpServlet {

    protected void processRequest(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");
        try (PrintWriter out = response.getWriter()) {
        }}
}
```

```

@Override
protected void doGet(HttpServletRequest request, HttpServletResponse response)
    throws ServletException, IOException {
    PrintWriter out= response.getWriter();
    String s1 = request.getParameter("n1");
    String s2 = request.getParameter("n2");
    String s3 = request.getParameter("n3");
    String s4 = request.getParameter("n4");
    try{
        Class.forName("com.mysql.jdbc.Driver");
        Connection con=
DriverManager.getConnection("jdbc:derby://localhost:1527/registester","root","root");
        String sql= "Insert into register values(?,?,?,?)";
        PreparedStatement ps = con.prepareStatement(sql);
        ps.setString(1, s1);
        ps.setString(2, s2);
        ps.setString(3, s3);
        ps.setString(4, s4);
        ps.executeUpdate();
        out.println("Data Inserted!");
        con.close();
    }
    catch(Exception e){
        out.println(e);
    }
}
}

```



Registration Servlet

localhost:8080/practical1c/

Enter Username:

Enter password:

Enter Email:

Enter Country Name:

Practical 2

A]Using Request Dispatcher Interface create a Servlet which will validate the password entered by the user, if the user has entered “Servlet” as password, then he will be forwarded to Welcome Servlet else the user will stay on the index.html page and an error message will be displayed.

Code:

index.html

```
<html>
  <head>
    <title>Request Dispatcher</title>
  </head>
  <body>
    <form action = "Demo" method = "get">
      Enter Username <input type = "text" name = "Username"><br><br>
      Enter Password <input type = "password" name = "Password">
      <input type = "submit" value = "Submit">
    </form>
  </body>
</html>
```

Demo.java

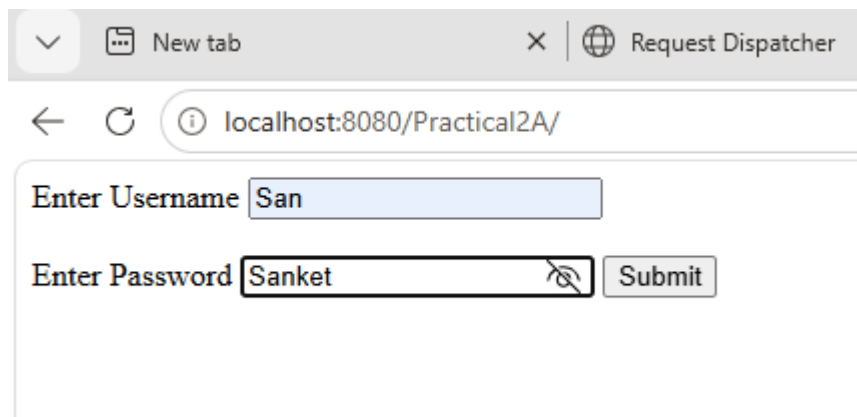
```
import java.io.*;
import javax.servlet.*;

public class Demo extends GenericServlet{
  public void service (ServletRequest request, ServletResponse response)
    throws ServletException, IOException {
    PrintWriter out = response.getWriter();
    String Username = request.getParameter("Username");
    String Password = request.getParameter("Password");
    if(Password.equals("Sanket")){
      request.setAttribute("User",Username);
      RequestDispatcher rd =
        request.getRequestDispatcher("Welcome");
      rd.forward(request,response);
    }else{
      out.println("Password is Invalid");
    }
  }
}
```

Welcome.java

```
import java.io.*;
import javax.servlet.*;

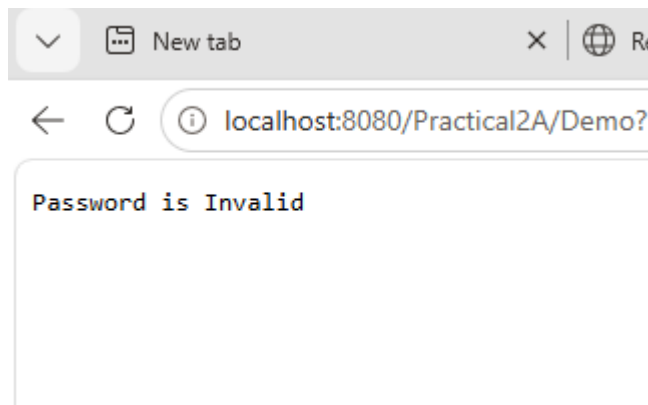
public class Welcome extends GenericServlet {
    public void service(ServletRequest request, ServletResponse response)
        throws ServletException, IOException {
        PrintWriter out = response.getWriter();
        String s = (String) request.getAttribute("San");
        out.println("Hello " + s + ", You are welcome here.");
    }
}
```



A screenshot of a web browser window. The address bar shows 'localhost:8080/Practical2A/'. The page contains a form with two input fields: 'Enter Username' with the value 'San' and 'Enter Password' with the value 'Sanket'. There is a 'Submit' button next to the password field. The browser's tab is labeled 'New tab' and the page title is 'Request Dispatcher'.



A screenshot of a web browser window. The address bar shows 'localhost:8080/Practical2A/'. The page displays the text 'Hello Sanket, You are welcome here.' in a monospaced font. The browser's tab is labeled 'New tab' and the page title is 'Request Dispatcher'.



A screenshot of a web browser window. The address bar shows 'localhost:8080/Practical2A/Demo?'. The page displays the text 'Password is Invalid' in a monospaced font. The browser's tab is labeled 'New tab' and the page title is 'Request Dispatcher'.

B]Create a servlet that uses Cookies to store the number of times a user has visited the servlet.

Code:

Index.html

```
<html>
  <head>
    <title>Cookie</title>
  <body>
    <form action = "Cookies" method="doGet">
      <input type="submit" value="Go to Servlet for visit counter">
    </form>
  </body>
</head>
</html>
```

Cookies.java

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class Cookies extends HttpServlet {

    private int i = 1;

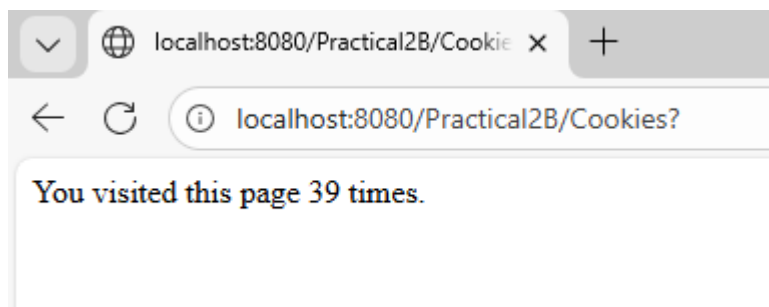
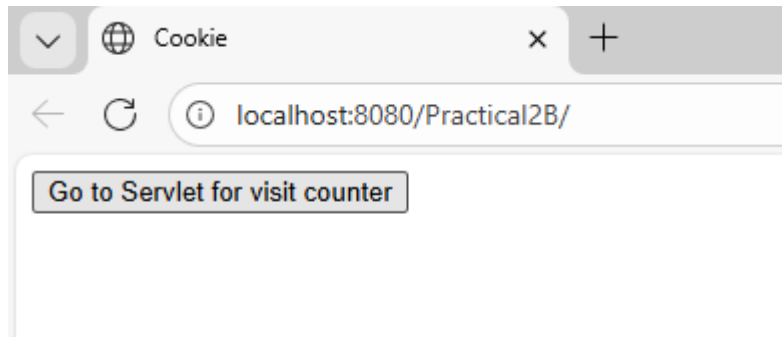
    public void doGet(HttpServletRequest req, HttpServletResponse res)
        throws ServletException, IOException {

        res.setContentType("text/html");
        PrintWriter out = res.getWriter();
        String k = String.valueOf(i);
        Cookie c = new Cookie("visit", k);
        res.addCookie(c);

        int j = Integer.parseInt(c.getValue());

        if (j == 1) {
            out.println("This is the first time you are visiting the page.");
        } else {
            synchronized (this) {
                out.println("You visited this page " + i + " times.");
            }
        }
    }
}
```

```
    i++;  
  }  
}
```



C]Create a servlet demonstrating the use of session creation and destruction. Also check whether the user has visited this page first time or has visited earlier also using sessions.

Code:

Index.html

```
<html>
  <head>
    <title></title>

  </head>
  <body>
    <form action="HttpsessionServlet" method="get">
      <input type="submit" value="Go to Servlet for session Demo"/>
    </form>
  </body>
</html>
```

HttpsessionServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpSession;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

public class HttpsessionServlet extends HttpServlet {
    private int counter;

    protected void doGet(HttpServletRequest request, HttpServletResponse response) throws
ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");
        PrintWriter out=response.getWriter();
        HttpSession session=request.getSession(true);
        if(session.isNew())
        {
            out.println("This is the first time you are visiting this page");
            ++counter;
        }else{
            synchronized(HttpsessionServlet.this)
            {
                if(counter==10)
```

$$\}$$

Practical 3

A]Create a Servlet application to upload and download a file.

Code(Upload):

Index.html

```
<!DOCTYPE html>
```

```
<!--
```

To change this license header, choose License Headers in Project Properties.

To change this template file, choose Tools | Templates

and open the template in the editor.

```
-->
```

```
<html>
```

```
  <head>
```

```
    <title>TODO supply a title</title>
```

```
    <meta charset="UTF-8">
```

```
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
  </head>
```

```
  <body>
```

```
    <form action="FileUploadServlet" method="post" enctype="multipart/form-data">
```

```
      Select file to Upload:<input type="file" name="file" id="file">
```

```
      Destination<input type="text" value="/tmp" name="destination">
```

```
      <br>
```

```
      <input type="submit" value="Upload file" name="upload" id="upload">
```

```
    </form>
```

```
  </body>
```

```
</html>
```

FileUploadServlet.java

```
import java.io.*;
```

```
import javax.servlet.*;
```

```
import javax.servlet.annotation.MultipartConfig;
```

```
import javax.servlet.http.*;
```

```
@MultipartConfig
```

```
public class FileUploadServlet extends HttpServlet {
```

```
    public void doPost(HttpServletRequest req, HttpServletResponse res) throws ServletException, IOException
```

```
    {
```

```
        res.setContentType("text/html");
```

```
        PrintWriter out=res.getWriter();
```

```
        String path=req.getParameter("destination");
```

```
        Part filePart=req.getPart("file");
```

```
        String sfilePart=req.getPart("file").toString();
```

```
        out.print("<br> FilePart: "+sfilePart);
```

```
        String filename=filePart.getSubmittedFileName().toString();
```

```
        out.print("<br><br><hr> file name: "+filename);
```

```

OutputStream os=null;
InputStream is=null;
try{
os=new FileOutputStream(new File(path+File.separator+filename));
is=filePart.getInputStream();
int read=0;
byte[] b=new byte[1024];
    while ((read=is.read(b))!=-1){
os.write(b, 0, read);
    }
    out.println("<br> file uploaded sucessfully...!");
}
catch(FileNotFoundException e){out.print(e);}
}}

```

Select file to Upload: msc it part 2 .pdf

Destination

FilePart: File name=msc it part 2 .pdf,
StoreLocation=C:\Users\ADMIN\AppData\Roaming\NetBeans\8.2\config\GF_4.1.1\domain1\generated\jsp\pract4c\upload__1f289f3c_19fd6113674__7ffb_00000003.tmp, size=9116678bytes,
isFormField=false, FieldName=file

file name: msc it part 2 .pdf
file uploaded sucessfully...!

Code (Download):

index.html

```

<html>
  <head>
    <title>Upload File</title>
  </head>
  <body>
    <h1>File Download Application</h1>
    Click<a href="DownloadServlet?Filename=Chapter-1.pdf">Sample Chapter</a>
  <br/><br/>
  </body>
</html>

```

DownloadServlet.java

```

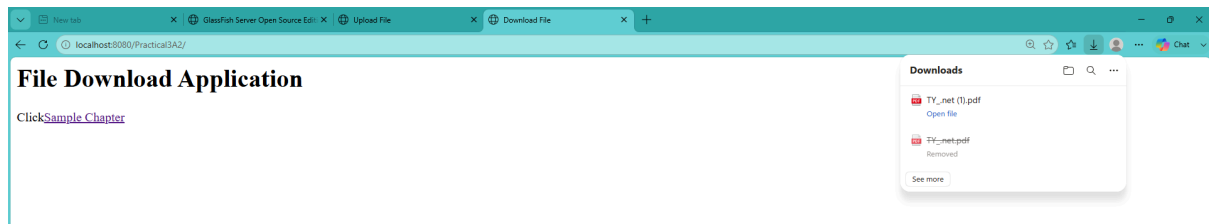
import java.io.*;
import javax.servlet.*;

```

```

import javax.servlet.http.*;
public class DownloadServlet extends HttpServlet {
    @Override
    public void doGet(HttpServletRequest request, HttpServletResponse response) throws
ServletException, IOException{
        PrintWriter out = response.getWriter();
        String filename ="TY_.net.pdf";
        String
filepath="//C://Users//Admin//OneDrive//Documents//NetBeansProjects//Practical3A2//web//
WEB-INF//";
        response.setHeader("Content-Disposition","attachment; filename="+filename);
        FileInputStream f=new FileInputStream(filepath+filename);
        int i;
        while((i=f.read())!=-1){
            out.write(i);
        }
        f.close();
    }
}

```



Practical 4

A]Develop a simple JSP application to display values obtained from the use of intrinsic objects of various types.

Code:

index.html

```
<html>
<body>
<form action="implicitObjectEx.jsp">
Enter your name:<input type="text" name="myname"><br><br>
Enter your email id:<input type="text" name="mymailid"><br><br>
<input type="submit" value="submit">
</form>
</body>
</html>
```

implicitObjectEx.jsp

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<html>
<head>
<title>JSP Page</title>
</head>
<body>
<h1>Use of Intrinsic Objects in JSP</h1>
<h1>Request Object</h1>
Query String<%=request.getQueryString() %><br> Context
Path<%=request.getContextPath()
%><br> Remote
Host<%=request.getRemoteHost()
%><br>
<h1>Response Object</h1> Character Encoding
Type<%=response.getCharacterEncoding() %><br> Content Type
<%=response.getContentType() %><br> Locale <%=response.getLocale() %><br>
<h1>Session Object</h1> ID<%=session.getId() %><br>
Creation Time<%=new java.util.Date(session.getCreationTime()) %><br>
Last Access Time<%=new java.util.Date(session.getLastAccessedTime()) %><br>
</body>
</html>
```

Use of Intrinsic Objects in JSP

Request Object

Query String myname=Sanket&mymailid=sanlsha0317%40gmail.com
Context Path /jsp_implicit
Remote Host 0:0:0:0:0:0:1

Response Object

Character Encoding Type UTF-8
Content Type text/html; charset=UTF-8
Locale en_IN
h1>Session Object ID 1f7ff02178426ac7a50f7cc2d295
Creation Time Thu Jul 30 13:10:47 IST 2026
Last Access Time Thu Jul 30 13:10:47 IST 2026

BJDevelop a simple JSP application to pass values from one page to another with validations. (Name-txt, age-txt, hobbies-checkbox, email-txt, gender-radio button).

Code:

Index.jsp

```
<%@ page language="java" contentType="text/html" pageEncoding="UTF-8"%>
<%!
int ageInNumbers;
private static final String EMAIL_REGEX =
"^[\\w-\\+](\\.[\\w]+)*@[\\w-](\\.[\\w]+)*(\\.[a-z]{2,})$";
%>
<%
String name = request.getParameter("fullname");
String age = request.getParameter("age");
String email = request.getParameter("email");
String gender = request.getParameter("gender");
String hb[] = request.getParameterValues("hob");
if (name.isEmpty() || age.isEmpty() || email.isEmpty() || gender.isEmpty())
{
    out.println("<font color=red>Please fill all the fields</font><br>" );
}
if (name== null | name.trim().isEmpty()){
    out.println("<h3 style='color:red'>Name cannot be empty.</h3>");
}
else if (!name.matches("[a-zA-Z ]+")) {
    out.println("<h3 style='color:red'>Name should contain only alphabets and spaces.</h3>");
} else {
    out.println("<h3 style='color:green'>Valid Name: " + name + "</h3>");
}
if (!email.matches(EMAIL_REGEX)) {
    out.println("<font color='red'>Correct Your Email Address</font><br>");
}
try {
    ageInNumbers = Integer.parseInt(age.trim());
} catch (NumberFormatException e) {
    out.println("<font color='red'>Age must be in numbers</font><br>");
}
if (ageInNumbers <= 18 && ageInNumbers >=60){
    out.println("<font color=red>Age must between 18 to 60</font><br>");
}
%>

<h1>Registration Form</h1>
```

Your Entered Information is as follows:

Full Name :<%=name%>

Age :<%=age%>

E-Mail :<%=email%>

Gender :<%=gender%>

Hobbies :

<%=

```
if(hb != null && hb.length != 0)
```

```
{
```

```
    for(int i = 0; i < hb.length; i++)
```

```
    {
```

```
        out.println(hb[i]);
```

```
    }
```

```
}
```

```
%>
```

index.jsp

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>
```

```
<html>
```

```
    <body>
```

```
        <h1>Registration Form</h1>
```

```
        <form action="val.jsp">
```

```
            Enter Your Name <input type="text" name="fullname"><br>
```

```
            Enter Your Age <input type="text" name="age"><br>
```

```
            Select Hobbies <input type="checkbox" name="hob" value="Singing"> Singing
```

```
            <input type="checkbox" name="hob" value="Reading">Reading Books
```

```
            <input type="checkbox" name="hob" value="Cricket">Playing Cricket<br>
```

```
            Enter E-mail<input type="text" name="email"><br>
```

```
            Select Gender <input type="radio" name="gender" value="Male">Male<br>
```

```
            <input type="radio" name="gender" value="female">Female<br>
```

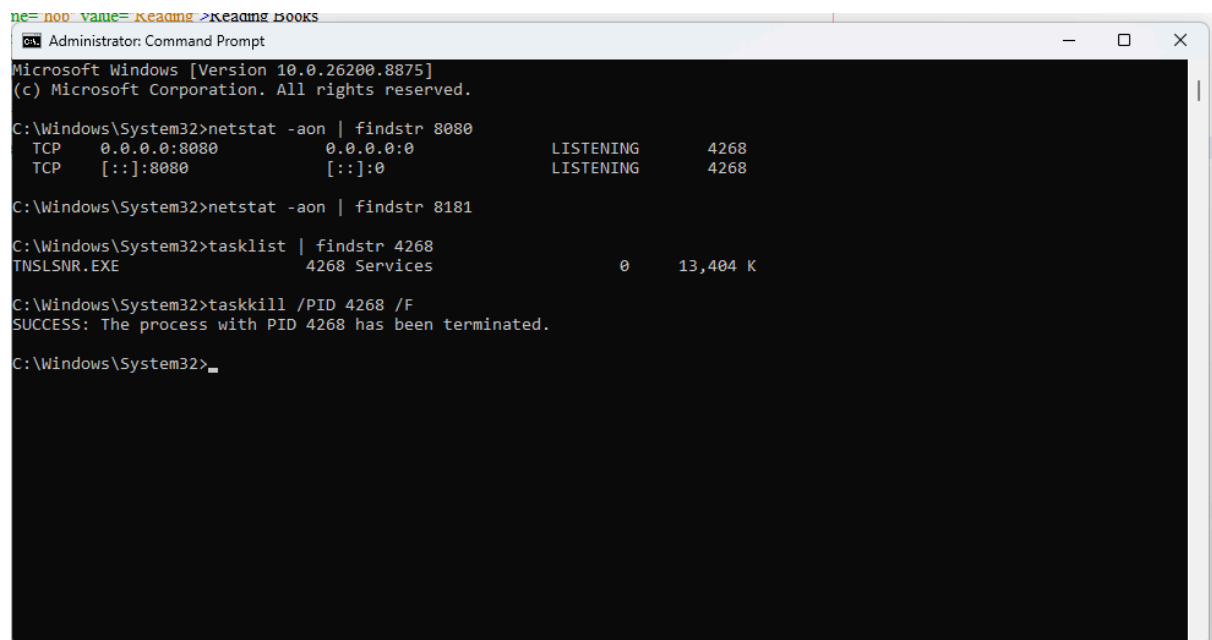
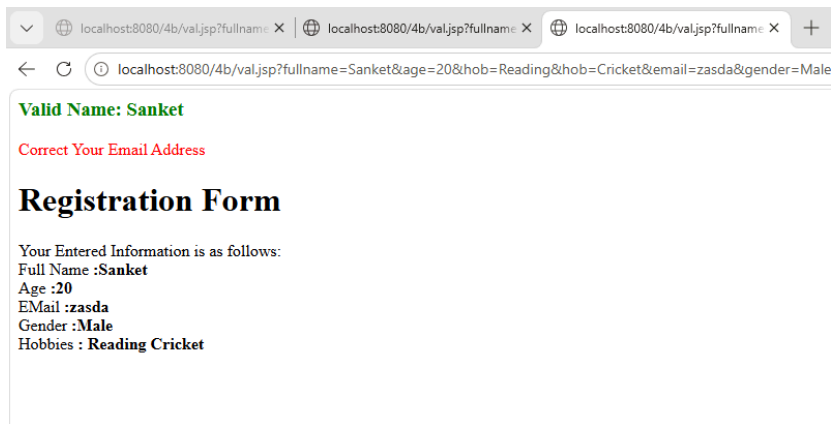
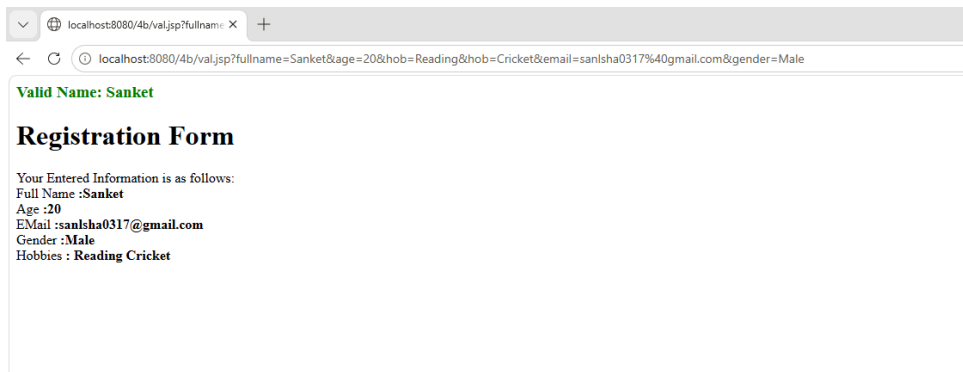
```
            <input type="radio" name="gender" value="Other">Other<br>
```

```
            <input type="Submit" value="submit Form">
```

```
        </form>
```

```
    </body>
```

```
</html>
```



C]Create a registration and login JSP application to register and authenticate the user based on username and password using JDBC.

Code:

Login.html

```
<html>
<body>
<h1>Login Page</h1>
<form action="Login.jsp">
Enter User Name<input type="text" name="txtName"><br>
Enter Password<input type="password" name="txtPass"><br>
<input type="submit" value="~~~LOGIN~~"><input type="reset">
</form>
</body>
</html>
```

index.html

```
<html>
<head>
<title>New User Registration Page</title>
</head>
<body>
<form action="Registration.jsp">
<h1>New User Registration Page</h1>
Enter User Name :<input type="text" name="txtName"><br><br>
Enter Password :<input type="password" name="txtPass1"><br><br>
Re-Enter Password :<input type="password" name="txtPass2"><br><br>
Enter Email :<input type="text" name="txtEmail"><br><br>
Enter Country Name :<select name="txtCon">
<option>India</option>
<option>France</option>
<option>England</option>
<option>Argentina</option>
</select><br>
<input type="submit" value="REGISTER"><input type="reset">
</form>
</body>
```

</html>

Login.jsp

```
<%@page contentType="text/html" import="java.sql.*"%>
<html><body>
<h1>Registration JSP Page</h1>
<%
String uname=request.getParameter("txtName");
String pass=request.getParameter("txtPass");
ResultSet rs=null;
try{
    // 1. Updated to Apache Derby driver
    Class.forName("org.apache.derby.jdbc.ClientDriver");

    // 2. Updated database name to userpass
    Connection
con=DriverManager.getConnection("jdbc:derby://localhost:1527/userpass","root","root");

    Statement stmt=con.createStatement();
    rs=stmt.executeQuery("select password from userpass where username='"+uname+"'");
    rs.next();
    if(pass.equals(rs.getString(1)))
    {
        out.println("<h1>~~~LOGIN SUCCESSFULL~~~</h1>");
    }
    else
    {
        out.println("<h1>password does not match!!!!</h1>");
        %>
        <jsp:include page="index.html"></jsp:include>
        <%
    }
} catch(Exception e){
    out.println("<h1>User does not exist!!!!</h1>");
    %>
    <jsp:include page="index.html"></jsp:include>
    <%
}
%>
```

```
</body>
</html>
```

Registration.jsp

```
<%@page contentType="text/html" import="java.sql.*"%>
<html><body>
<h1>Registration JSP Page</h1>
<%
String uname=request.getParameter("txtName");
String pass1=request.getParameter("txtPass1");
String pass2=request.getParameter("txtPass2");
String email=request.getParameter("txtEmail");
String ctry=request.getParameter("txtCon");
if(pass1.equals(pass2))
{
try
{

    Class.forName("org.apache.derby.jdbc.ClientDriver");
    Connection
con=DriverManager.getConnection("jdbc:derby://localhost:1527/userpass","root","root");
PreparedStatement stmt=con.prepareStatement("insert into userpass values(?,?,?,?)");
stmt.setString(1,uname);
stmt.setString(2,pass1);
stmt.setString(3,email);
stmt.setString(4,ctry);
int row=stmt.executeUpdate();
if(row==1)
{
out.println("Registration Successful");}
else
{
out.println("Registration FAILED!!!!");
}%>
<jsp:include page="index.html"></jsp:include>
<%
}
} catch(Exception e){out.println(e);}

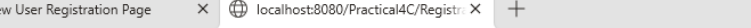
```

```
}
else
{
out.println("<h1>Password Mismatch</h1>");
%>
<jsp:include page="index.html"></jsp:include>
<% }
%>
</body>
</html>
```

The screenshot shows a web browser window with two tabs, both titled 'New User Registration Page'. The address bar displays 'localhost:8080/Practical4C/'. The page content includes a heading 'New User Registration Page' and a registration form. The form fields are as follows:

- Enter User Name :** A text input field containing the value 'Sanket'.
- Enter Password :** A password input field containing three asterisks '***'.
- Re-Enter Password :** A password input field containing three asterisks '***'.
- Enter Email :** A text input field containing the value 'sanlsha@gmail.com'.
- Enter Country Name :** A dropdown menu with 'India' selected.

At the bottom of the form, there are two buttons: 'REGISTER' and 'Reset'.



Registration JSP Page

Registration Successful

The screenshot displays a Windows desktop environment. On the left, a terminal window titled 'NetworkMiner.py 52' shows the output of a network scan. The scan identifies a host at 192.168.1.100 as a Linux system with various services running, including SSH, Telnet, and a web server. The scan results are as follows:

```

Host: 192.168.1.100 [Linux] (Ubuntu)
  Host is up (0.0000s)
    tcp: 22
    tcp: 23
    tcp: 80
    tcp: 443
    tcp: 8080
    tcp: 8443
    tcp: 9000
    tcp: 9001
    tcp: 9002
    tcp: 9003
    tcp: 9004
    tcp: 9005
    tcp: 9006
    tcp: 9007
    tcp: 9008
    tcp: 9009
    tcp: 9010
    tcp: 9011
    tcp: 9012
    tcp: 9013
    tcp: 9014
    tcp: 9015
    tcp: 9016
    tcp: 9017
    tcp: 9018
    tcp: 9019
    tcp: 9020
    tcp: 9021
    tcp: 9022
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    tcp: 9412
    tcp: 9413
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    tcp: 9418

```

Practical 6

A]Create an HTML page with fields, eno, salary. Now on submit this data to a JSP page which will update the employee table of database with matching eno.

Code:

Index.html

```
<html>
  <head>
    <title>Practical 4C</title>
  </head>
  <body>
    <form action="UpdateEmp.jsp">
      Employee Number<input type="text" name="txtEno">
      Employee Salary(Update)<input type="text" name="txtSal">
      <input type="Submit" value="Submit">
      <input type="reset" value="Reset">
    </form>
  </body>
</html>
```

UpdateEmp.jsp

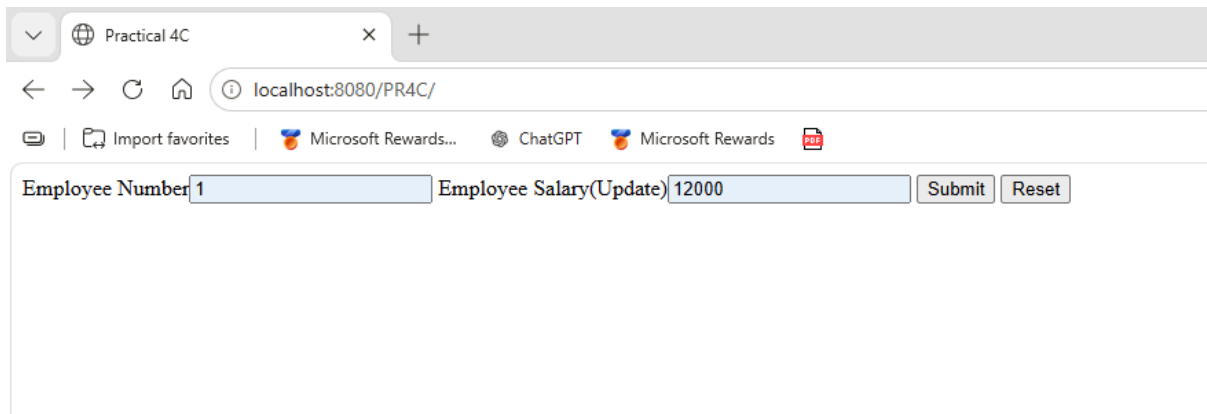
```
<%@page import="java.sql.*"%>
<%@page contentType="text/html" pageEncoding="UTF-8"%>

<html>
  <body>
    <h1>Updating Employee Record</h1>
    <%
      String eno=request.getParameter("txtEno");
      String sal=request.getParameter("txtSal");
      try{
        Class.forName("com.mysql.jdbc.Driver");
        Connection con =
DriverManager.getConnection("jdbc:derby://localhost:1527/Emp","Emp","root");
        PreparedStatement stmt=con.prepareStatement("select * from Emp where eno=?");
        stmt.setString(1, eno);
        ResultSet rs=stmt.executeQuery();
        if(rs.next()){
          out.println("<h1>Employee"+rs.getString(2)+"Exist</h1>");
          PreparedStatement pst=con.prepareStatement("update Emp set sal=? where eno=?");
          pst.setString(1, sal);
          pst.setString(2,eno);
```

```

        pst.executeUpdate();
        out.println("<h1>Employee Record Updated</h1>");
    }
    else{
        out.println("<h1>Employee already exists</h1>");
    }
}
catch(Exception e){out.println(e);}
%>
</body>
</html>

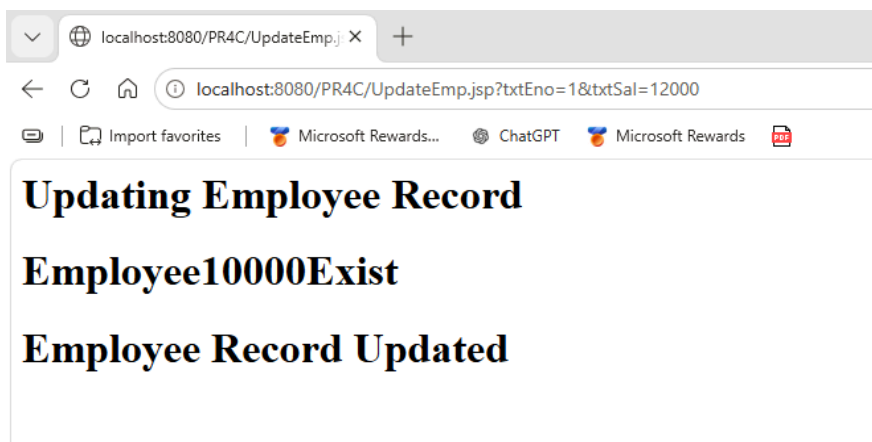
```



Practical 4C

localhost:8080/PR4C/

Employee Number Employee Salary(Update)



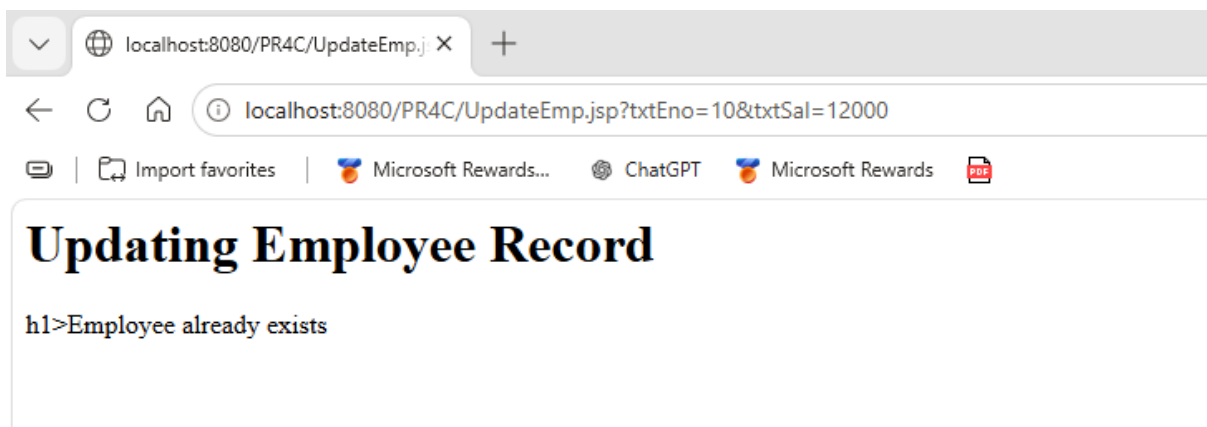
localhost:8080/PR4C/UpdateEmp.j

localhost:8080/PR4C/UpdateEmp.jsp?txtEno=1&txtSal=12000

Updating Employee Record

Employee10000Exist

Employee Record Updated



localhost:8080/PR4C/UpdateEmp.j

localhost:8080/PR4C/UpdateEmp.jsp?txtEno=10&txtSal=12000

Updating Employee Record

h1>Employee already exists

B]Create a JSP page to demonstrate the use of Expression language.

Code:

ELArithmeticOperator.jsp

```
<body>

    5*5+4: ${5*5+4} <br>

    1.4E4+1.4: ${1.4E4+1.4}<br>

    10 mod 4:${10 mod 4} <br>

    15 div 3: ${15 div 3}<br>

</body>
```

ELLogicalOperator.jsp

```
<body>

    <h2>Logical Operator</h2>

    true and true: ${true and true}<br>

    true && false: ${true && false}<br>

    true or true: ${true or true}<br>

    true || false: ${true || false}<br>

    not true: ${not true}<br>

    !false: ${!false}

</body>
```

ELRelationalOperator.jsp

```
<body>
```

<h2>Relational Operator</h2>

10.0==10: \${10.0==10}

10.0 eq 10: \${10.0 eq 10}

((20*10)!= 200): \${((20*10)!= 200)}

3 ne 3: \${3 ne 3}

3.2>=2: \${3.2>=2}

3.2 ge 2: \${3.2 ge 2}

2<3: \${2<3}

4 lt 6: \${4 lt 6}

2 <= 4: \${2 <= 4}

4 le 2: \${4 le 2}

</body>

ELconditional_op.jsp

<body>

<h2>Conditional Operator</h2>

The result of 10>2 is: "\${(10>1)?'greater':'lesser'}"

</body>

EmptyOperator.jsp

<body>

<H1>Empty Operator Example</H1>

The Value for the Empty operator is:: \${empty "txt"}

</body>

ExpressionLanguage.jsp

<body>

Welcome ,\${param.name}

Session Value is, \${sessionScope.user}

Cookie name is, \${cookie.name.value}

</body>

Index.jsp

<body>

<h3>Welcome to index page</h3>

<% session.setAttribute("user","Admin"); %>

<%

Cookie ck=new Cookie("name","mycookie");

response.addCookie(ck);

%>

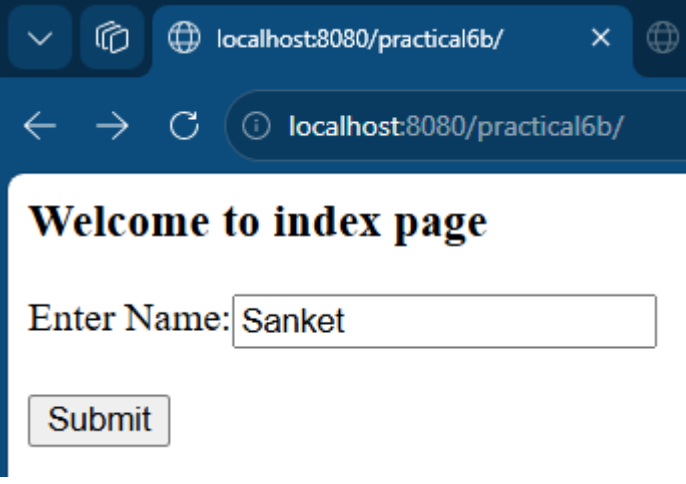
<form action="ExpressionLanguage.jsp">

Enter Name:<input type="text" name="name" />

<input type="submit" value="Submit"/>

</form>

</body>

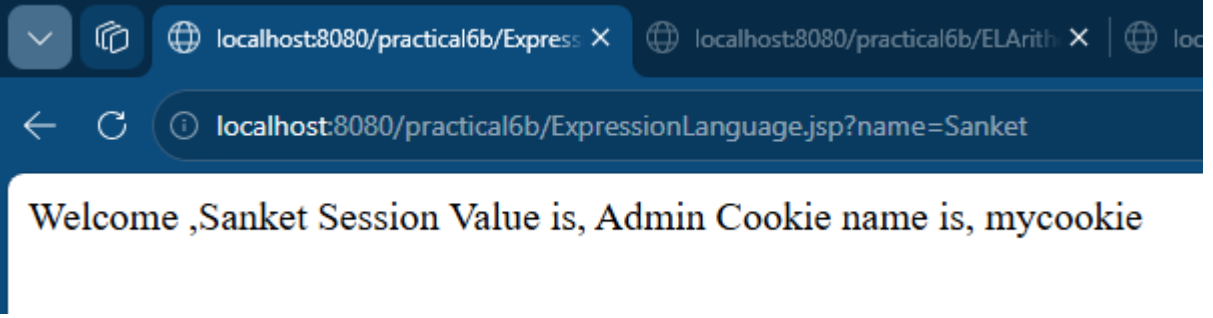


A screenshot of a web browser window. The address bar shows 'localhost:8080/practical6b/'. The page content includes a heading 'Welcome to index page', a text input field with the value 'Sanket', and a 'Submit' button.

localhost:8080/practical6b/

Welcome to index page

Enter Name:

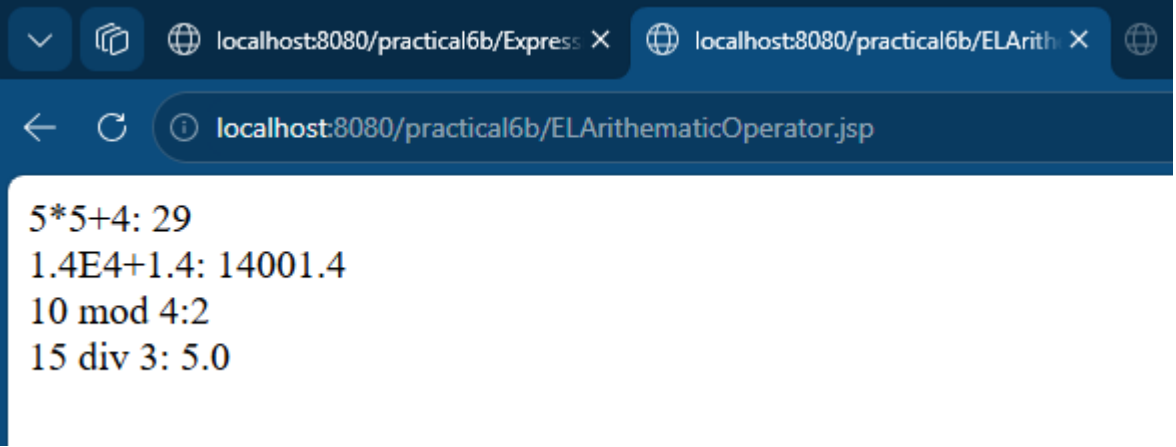


A screenshot of a web browser window. The address bar shows 'localhost:8080/practical6b/ExpressionLanguage.jsp?name=Sanket'. The page content displays a welcome message and session/cookie information.

localhost:8080/practical6b/Express X | localhost:8080/practical6b/ELArith X | loc

← ↻ ⓘ localhost:8080/practical6b/ExpressionLanguage.jsp?name=Sanket

Welcome ,Sanket Session Value is, Admin Cookie name is, mycookie

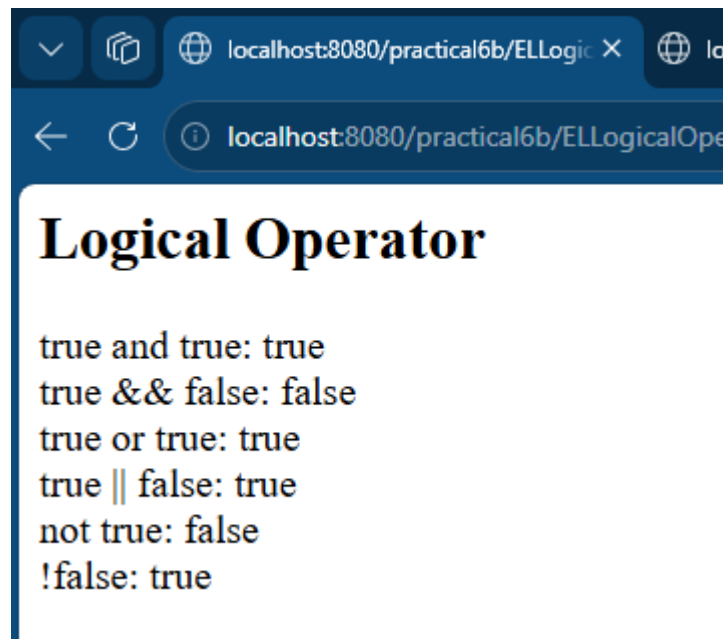


A screenshot of a web browser window. The address bar shows 'localhost:8080/practical6b/ELArithmeticOperator.jsp'. The page content displays the results of several arithmetic operations.

localhost:8080/practical6b/Express X | localhost:8080/practical6b/ELArith X | loc

← ↻ ⓘ localhost:8080/practical6b/ELArithmeticOperator.jsp

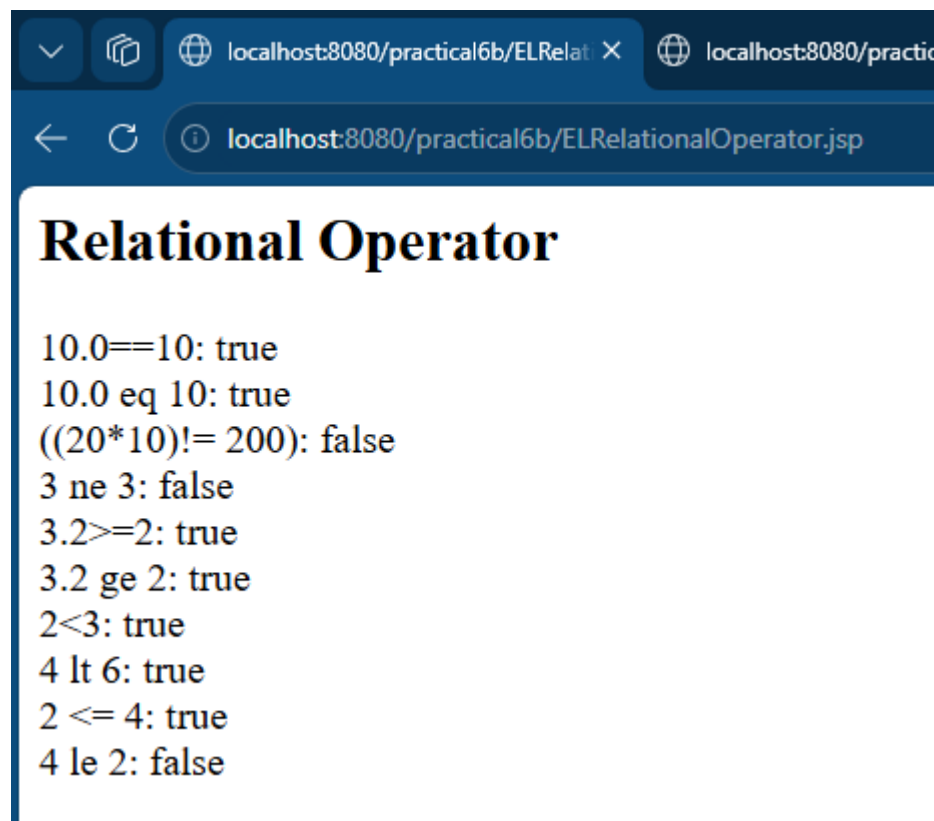
5*5+4: 29
1.4E4+1.4: 14001.4
10 mod 4:2
15 div 3: 5.0



localhost:8080/practical6b/ELLogic X localhost:8080/practical6b/ELLogicalOperator.jsp

Logical Operator

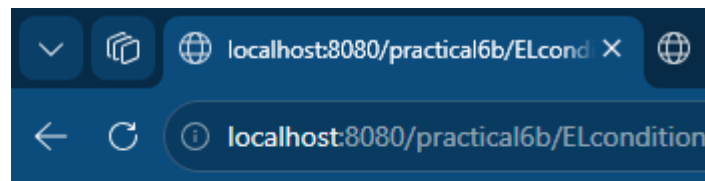
true and true: true
true && false: false
true or true: true
true || false: true
not true: false
!false: true



localhost:8080/practical6b/ELRelat X localhost:8080/practical6b/ELRelationalOperator.jsp

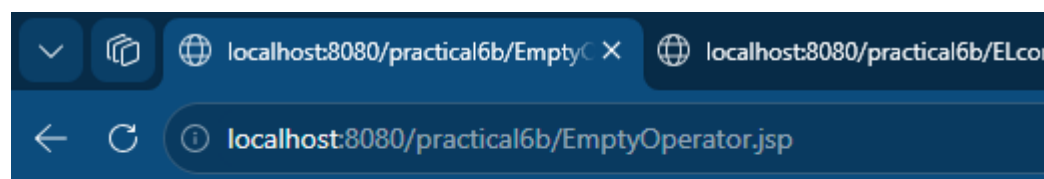
Relational Operator

10.0==10: true
10.0 eq 10: true
((20*10)!= 200): false
3 ne 3: false
3.2>=2: true
3.2 ge 2: true
2<3: true
4 lt 6: true
2 <= 4: true
4 le 2: false



Conditional Operator

The result of $10 > 2$ is: "greater"



Empty Operator Example

The Value for the Empty operator is:: false

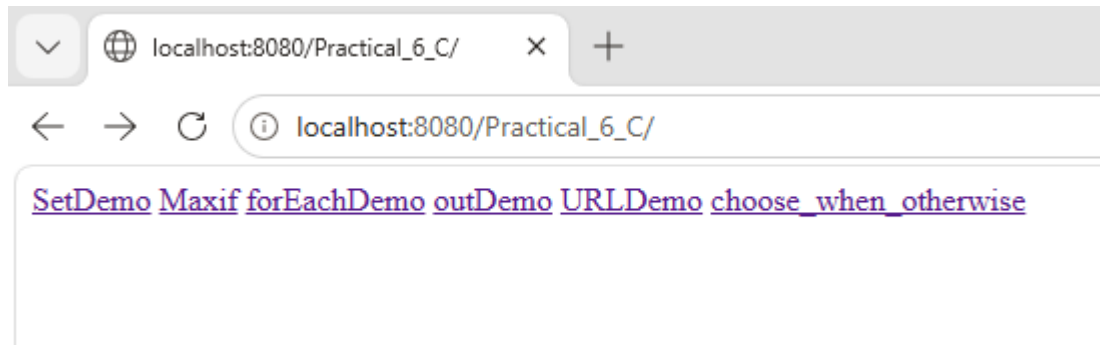
C]Create a JSP application to demonstrate the use of JSTL.

Code:

Code:

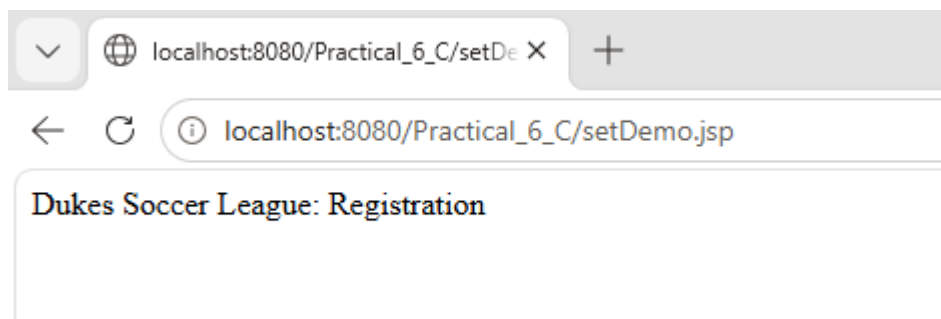
Index.html

```
<html>
  <body>
    <a href="setDemo.jsp">SetDemo</a>
    <a href="Maxif.html">Maxif</a>
    <a href="forEachDemo.jsp">forEachDemo</a>
    <a href="NameFile.html">outDemo</a>
    <a href="URLDemo.jsp">URLDemo</a>
    <a href="choose_when_otherwise.jsp">choose_when_otherwise</a>
  </body>
</html>
```



setDemo.jsp

```
<%@taglib prefix="c" uri="http://java.sun.com/jstl/core"%>
<c:set var="pageTitle" scope="application"
value="Dukes Soccer League: Registration" />
${pageTitle}
```



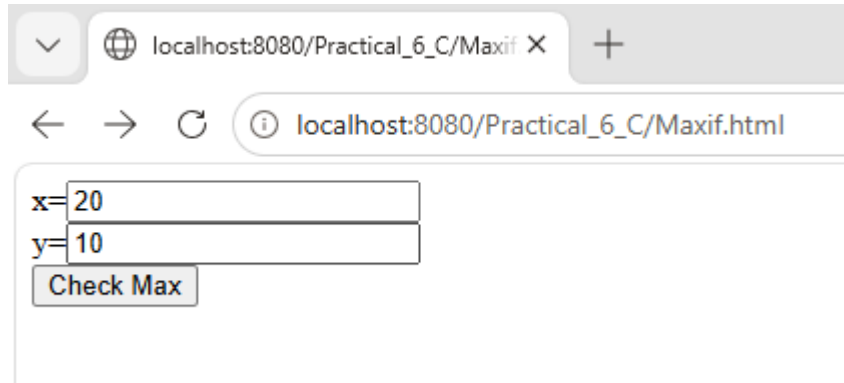
Maxif.html

```
<html>
  <body>
    <form action="IFDemo.jsp">
```

```

        x=<input type="text" name="x"/><br>
        y=<input type="text" name="y"/><br>
        <input type="submit" value="Check Max"/>
    </form>
</body>
</html>

```

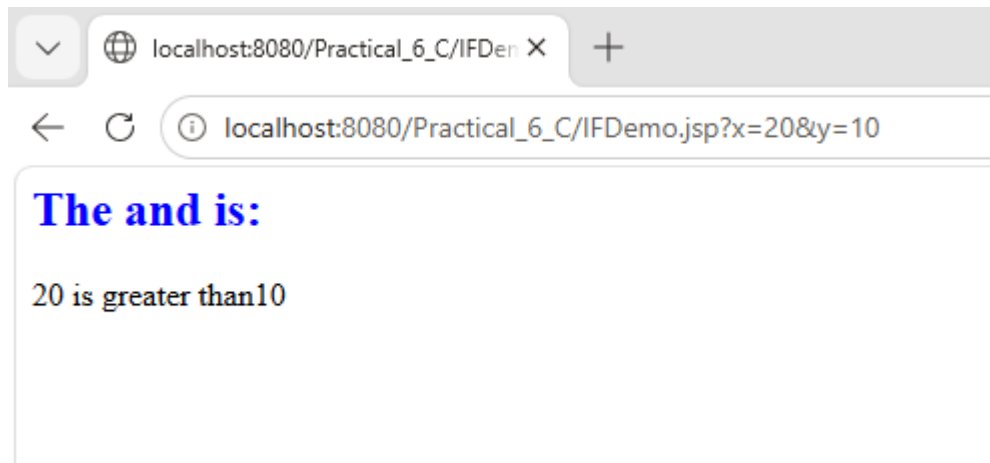


A screenshot of a web browser window. The address bar shows 'localhost:8080/Practical_6_C/Maxif.html'. The page content displays a form with two text input fields. The first field is labeled 'x=' and contains the number '20'. The second field is labeled 'y=' and contains the number '10'. Below these fields is a button labeled 'Check Max'.

```

IFDemo.jsp
<%@taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core"%>
<c:set var="x" value="{param.x}"/>
<c:set var="y" value="{param.y}"/>
<c:if test="{x}>{y}">
    <font color="blue"><h2>The and is:</h2></font>
    <c:out value="{x} is greater than${y}"/>
</c:if>

```

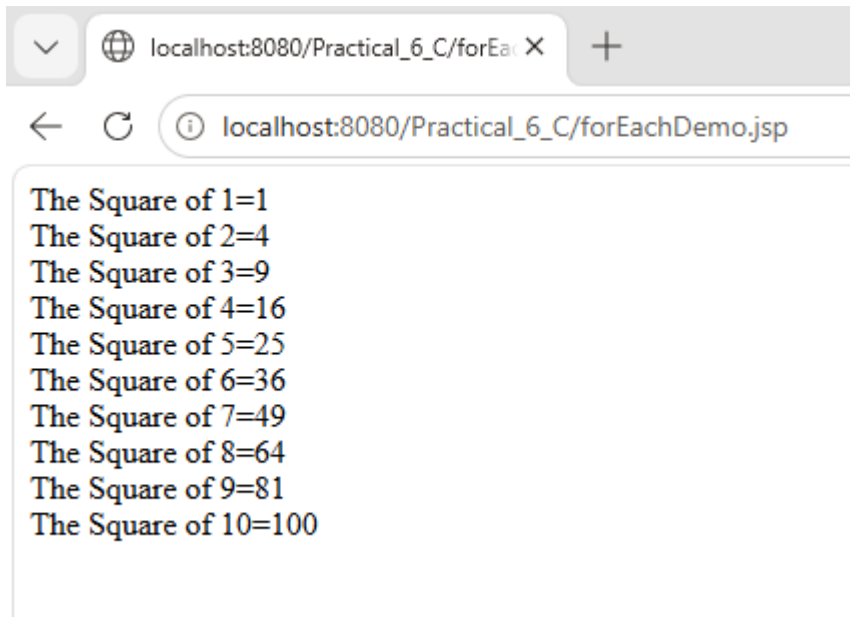


A screenshot of a web browser window. The address bar shows 'localhost:8080/Practical_6_C/IFDemo.jsp?x=20&y=10'. The page content displays the text 'The and is:' in blue, followed by '20 is greater than10'.

```

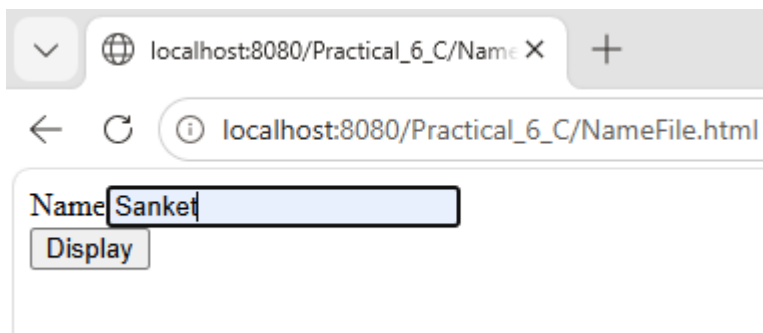
forEachDemo.jsp
<%@taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core"%>
<c:forEach begin="1" end="10" var="i">
    The Square of <c:out value="{i}=${i*i}"/><br>
</c:forEach>

```



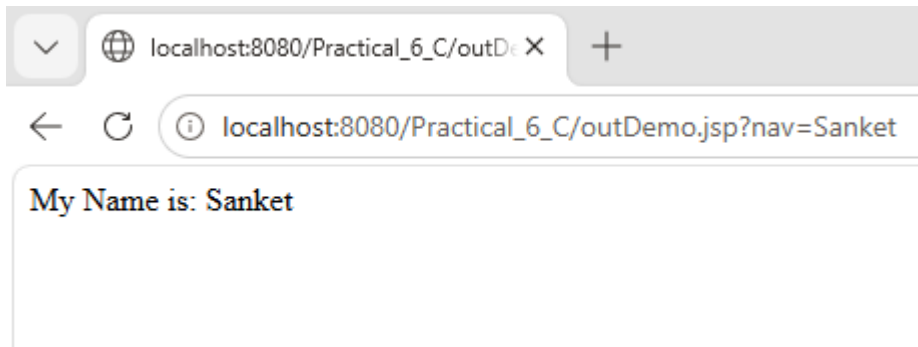
NameFile.html

```
<html>
  <body>
    <form action="outDemo.jsp">
      Name<input type="text" name="nav"/><br>
      <input type="submit" value="Display"/>
    </form>
  </body>
</html>
```



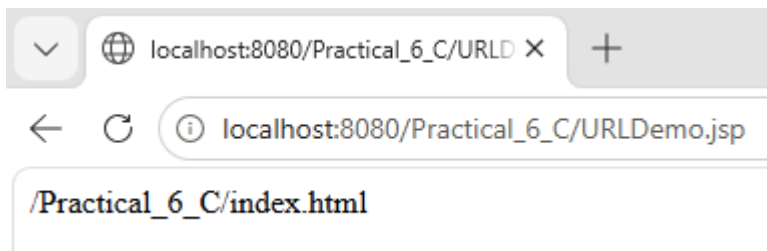
outDemo.jsp

```
<%@taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core" %>
<:set var="nav" value="${param.nav}"/>
  My Name is: <c:out value="${nav}"/>
```



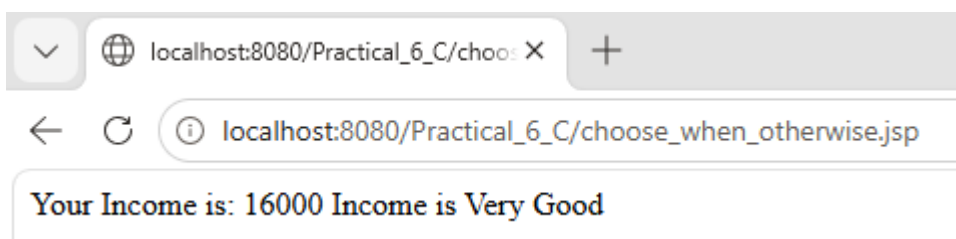
URLDemo.jsp

```
<%@taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core" %>
<c:url value="/index.html"/>
```



Choose_when_otherwise.jsp

```
<%@taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core" %> <c:set var="income"
value="${4000*4}"/>
Your Income is: <c:out value="${income}"/>
<c:choose>
  <c:when test="${income <=1000}">
    Income is not good
  </c:when>
  <c:when test="${income > 10000}">
    Income is Very Good
  </c:when>
  <c:otherwise>
    Income is undetermined
  </c:otherwise>
</c:choose>
```



Practical 7

a) Create a Currency Converter application using EJB.

Index.html

Code:

```
<html><head><title>Currency Converter</title></head>
<body>
<form action="CCServlet" >
Enter Amount <input type="text" name="amt"><br>
Select Conversion Type
<input type="radio" name="type" value="r2d" checked>Rupees to Dollar
<input type="radio" name="type" value="d2r" >Dollar to Rupees<br>
<input type="reset" ><input type="submit" value="CONVERT" >
</form>
</body>
</html>
```

CCBeanLocal.java

Code:

```
package mybeans;
//import javax.ejb.Stateless;
import javax.ejb.Local;
//@Stateless
@Local
public interface CCBeanLocal {
public double r2Dollar(double r);
public double d2Rupees(double d);
}
```

CCBean.java

Code:

```
package mybeans;
import javax.ejb.Stateless;
@Stateless
public class CCBean implements CCBeanLocal
{
public double r2Dollar(double r)
{
return r/95.51;
}
public double d2Rupees(double d)
{
return d*95.51;
}
```

```
}  
}
```

CCServlet.java

Code:

```
package mypack;  
import java.io.*;  
import javax.servlet.*;  
import javax.servlet.http.*;  
import javax.ejb.EJB;  
import mybeans.CCBeanLocal;  
public class CCServlet extends HttpServlet{  
    @EJB CCBeanLocal obj;  
    public void doGet(HttpServletRequest request, HttpServletResponse response)throws  
        ServletException, IOException  
    {  
        response.setContentType("text/html;charset=UTF-8");  
        PrintWriter out = response.getWriter();  
        double amt=Double.parseDouble(request.getParameter("amt"));  
        if(request.getParameter("type").equals("r2d")){  
            out.println("<h1>"+amt+"Rupees="+obj.r2Dollar(amt)+"Dollars</h1>");  
        }  
        if(request.getParameter("type").equals("d2r")){  
            out.println("<h1>"+amt+"Dollars="+obj.d2Rupees(amt)+"Rupees</h1>");  
        }  
    }  
}
```

Currency Converter

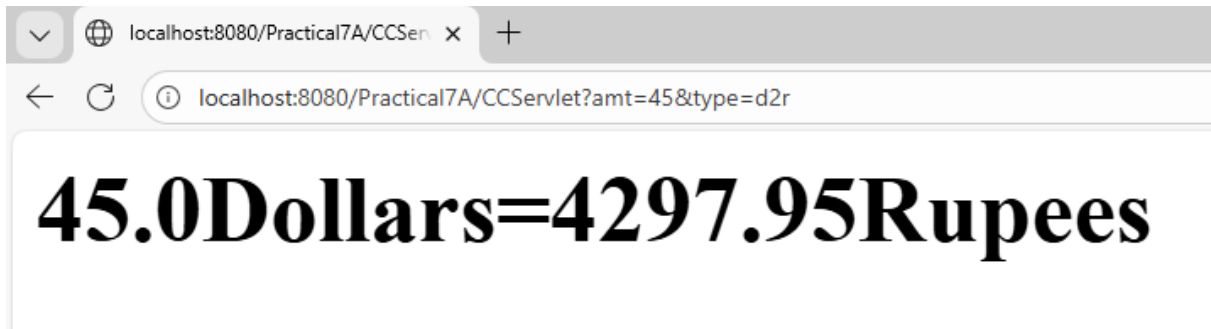
localhost:8080/Practical7A/

Enter Amount

Select Conversion Type ☒ Rupees to Dollar ☐ Dollar to Rupees

localhost:8080/Practical7A/CCServlet?amt=45&type=r2d

45.0Rupees=0.47115485289498477Dollars



BJDevelop a simple room reservation system application using EJB.

Code:

Index.html

```
<html>
```

```
<head><title>Room Reservation</title></head>
```

```
<body>
```

```
<p><b>Enter Details For Room Reservation.</b></p>
```

```
<form method="get" action="ReservationServlet">
```

```
Customers Name : <input type="text" name="cname" value="" size="20" /><br>
```

```
Address : <input type="text" name="cadd" value="" size="20" /><br><br>
```

```
Phone Number : <input type="text" name="cph" value="" size="12" /><br><br>
```

Room Type :

```
<select name="roomtype">
  <option id="g">General</option>
  <option id="d">Deluxe</option>
  <option id="s">Suite</option>
</select>
```

```
<br><br><br><br>
```

Check IN Date (DD/MM/YYYY):

```
<input type="text" name="checkindate" /><br>
```

Check Out Date (DD/MM/YYYY) :

```
<input type="text" name="checkoutdate" /><br>
```

Payment Mode :

```
<select name="paymode">
  <option id="cash">CASH</option>
  <option id="credit">CREDIT CARD</option>
</select>
```

```
<br><br><br>
```

```
<input name="cmdSubmit" type="submit" value="Submit" alt="Convert" />
```

```
</form>
```

```
</body>
```

```
</html>
```

ReservationBean.java

```
package servlet;
import EJPracts.ReservationBean;
import java.io.IOException;
import java.io.PrintWriter;
import java.util.ArrayList;
import javax.ejb.EJB;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
public class ReservationServlet extends HttpServlet {
    @EJB
    ReservationBean rb;
    protected void doGet (HttpServletRequest req, HttpServletResponse res) throws
    ServletException, IOException{
        res.setContentType("text/html;charset=UTF-8");
        String cname=req.getParameter("cname");
        String add=req.getParameter("cadd");
        String ph=req.getParameter("cph");
        String roomtype=req.getParameter("roomtype");
        String chkin=req.getParameter("checkindate");
        String chkout=req.getParameter("checkoutdate");
        String paymode=req.getParameter("paymode");
        PrintWriter out =res.getWriter();
        rb=new ReservationBean();
        int i = 0;
        ArrayList<String>list1=rb.reserve(cname, add, ph, chkin, chkout, roomtype, paymode);
        try{
            rb.reserve(cname,add,ph,chkin,chkout,roomtype,paymode);
            out.println("<html>");
            out.println("<head><title>ReservationEJB</title></head>");
            out.println("body");
            out.println(rb.welcome(cname));
            out.println(rb.roomType(roomtype));
            out.print(rb.payment(roomtype));
```

```

out.println("Your Details:");
int n =list1.size();
out.println("<Table Border=1>");
out.println("<tr><th>Name</th><th>Address</th><th>Phone</th><th>Check in
Date</th>");
out.println("<th>Check Out Drive</th><th>Room</th><th>Payment Mode</th></tr>0");
out.println("<tr>");
for(i=0; i<n; i++){
out.println("<TD>");
out.println(list1.get(i));
out.println("</TD>");
}
out.println("</tr>");
out.println("</TABLE>");
out.println("<br/>Click <a href='index,html'>here<a/> to go back");
out.println("</body>");
out.print("</html>");

} finally{
out.close();
}
}
}
}

```

ReservationServlet.java

```

package EJPracts;
import java.util.ArrayList;
import javax.ejb.Stateless;
@Stateless
public class ReservationBean{
public String welcome(String customer){
return "Hello"+" "+customer.toUpperCase()+" "+"Welcome to our hotel";
}
public String roomType(String roomtype ){
return "Hello"+" "+roomtype.toUpperCase()+" "+"Room";
}
public String payment(String roomtype){
System.out.print(roomtype);
System.out.print(roomtype);
if(roomtype.equalsIgnoreCase("SUITE"))
{
return "You have to pay 8000";
}
}
}

```

```

        else if(roomtype.equalsIgnoreCase("Duluxe")){
            return "You have to pay 4000";
        }
        else{
            return "You have to pay 2000";
        }
    }
    public ArrayList<String>reserve(String customer, String add, String ph, String checkin,
String checkout, String roomtype, String paymode )
    {
        ArrayList<String> a = new ArrayList();
        a.add(customer);
        a.add(add);
        a.add(ph);
        a.add(checkin);
        a.add(checkout);
        a.add(roomtype);
        a.add(paymode);
        return a;
    }
}

```

Room Reservation

Customers Name:

Address:

Phone Number Room Type: General Deluxe Suite

CheckinDate(DD/MM/YYYY):

CheckoutDate(DD/MM/YYYY):

Payment Mode: Cash Online

Enter Details For Room Reservation.

Customer Name :

Address :

Phone Number :

Room Type :

Check IN Date (DD/MM/YYYY) :

Check Out Date (DD/MM/YYYY) :

Payment Mode :

localhost:3000/practice/reservation?details={name:sanket patil,address:B-wing 206 building No.1 kr,phone:1234567890,checkin:01-09-2026,checkout:07-09-2026,room:General,payment:CASH}

Hello SANKET PATIL Welcome to Our Hotel

You have Selected GENERAL Room

You have to pay 2000 Rs

Your Details :

Name	Address	Phone	Check In Date	Check Out Date	Room	Payment Mode
sanket patil	B-wing 206 building No.1 krishna park pale budruk	1234567890	2026-09-01	2026-09-07	General	CASH

Click [here](#) to go back

Practical 8

A] Develop simple EJB application to demonstrate Servlet Hit count using Singleton Session Beans.

Code:

Index.html

```
<html>
<head>
<title>TODO supply a title</title>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<meta http-equiv="Refresh" content="0; URL=ServletClient">
</head>
<body>
<div>TODO write content</div>
</body>
</html>
```

CountServletHitsBean.java

```
package ejb;
import javax.ejb.Singleton;
@Singleton
public class CountServletHitsBean {
    private int hitCount;
    public synchronized int getCount()
    {
        return hitCount++;
    }
}
```

ServletClient.java

```
package servlet;
import ejb.CountServletHitsBean;
import java.io.*;
import javax.ejb.EJB;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.*;
@WebServlet(name = "ServletClient", urlPatterns = {"/ServletClient"})
public class ServletClient extends HttpServlet {
    @EJB CountServletHitsBean obj;
    @Override
    protected void service(HttpServletRequest req, HttpServletResponse res)throws
    ServletException, IOException{
```

```
res.setContentType("text/html");  
PrintWriter out=res.getWriter();  
out.println("<b>Number of times this Servlet is accessed</b>:"+obj.getCount());  
}  
}
```

